

The reliable bestseller in the 2 MW class.
The Senvion MM series.



Our portfolio, our outcome, our service for your success.



Our modular product portfolio matches your local requirements.

Our technology has to prove itself every day under real-world conditions. Thanks to our diverse product portfolio, we are able to answer your project needs, handle specific influences such as hot or cold temperatures, fulfill challenging sound requirements, and match high expectations in terms of grid connectivity. With the right configuration from one of the largest product portfolios on the market, you can count on a stable income – even with unstable project conditions.



Our proven outcome secures your profitable income.

Our company relies on more than 25 years of experience, and throughout that time we have been able to adapt our tried-and-tested concepts to new developments. Our proven evolutionary design approach can be seen in the high availability of our turbine fleet and its reliable and robust performance. To make sure that you get the security you need for your project planning, we allow our power performance to be validated by the independent advisor DNV GL. This evidence successfully demonstrates the accuracy of our power curve. With us as your experienced partner, you can put even your complex projects onto a sustainable and profitable path.



Our comprehensive service solutions ensure your turbine availability.

The high reliability of our turbines is the result of farsighted planning. Well-thought-out constructions enable access to all parts – even to the hub. Furthermore, our 700 service employees – part of a nationwide network of easily reachable service centers – care about seamless availability. Our conservative design approach allows them to operate effectively and safely, no matter how stable the conditions may be.

We offer more than 25 years of experience and 24-hour remote monitoring to match your individual requirements.

Experience is the best foundation.
7,317 wind turbine installations.



Over 25 years in the business, more than 7,317 wind turbine installations worldwide with a combined output of more than 15.5 gigawatts and uncompromising engineering skills have all earned Senvion one of the best reputations in the wind energy industry. We are highly ambitious: we constantly aim to improve not only our own turbines, but also to set new standards for the entire market.

Our answer to your challenge

We develop, produce and supply wind turbines to suit almost any location, with a rated power ranging from 2 to 6.2 megawatts. A quarter of a century of pioneering work has taught us that the more energy we invest in our work, the higher the yield for you. We offer a wide range of onshore wind turbines with different hub heights, ensuring we can deliver the best possible results for you in any wind class.

Our expertise for your project

Whether a single wind turbine or a whole wind farm, we take a personal interest in your project and keep an eye on every detail right from the start. We can offer customised solutions for blade production, foundation design, transport, installation, service and maintenance for your individual project. Our engineers are more than willing to listen to your suggestions and come up with the ideal solution in consultation with you, whilst paying due attention to the requirements of the market concerned.



Intelligent technology for safe operations.



"Senvion has been a pleasure to work with at Oakdale. We have found the team to be very responsive in its approach to PfR as a new customer and the technical support has been excellent throughout the construction and commissioning processes."

Simon Vince,
Construction and Operations Director for PfR



Our wind turbines have to cope with highly diverse requirements and operate reliably and safely even in extreme climate conditions, in difficult terrain or densely populated regions. To ensure that they do so, they come with intelligent technologies:

- **Innovative blade design:** in-house development and production of the rotor blades improves rotor aerodynamics and enables high energy yield combined with low noise emission.
- **Highly developed pitch system:** DC electric motors allow the blades to be pitched quickly and efficiently. This delivers a high energy yield in all operating conditions and also permits a controlled shutdown at any time – even during a power failure, thanks to battery backup.
- **Reliable lightning protection:** our wind turbines are equipped with internal and external lightning protection according to IEC protection class I.
- **Optional extras:** noise restrictions, shadow flicker, bat protection and other aspects all mean that your wind turbines have to be adjusted to suit your particular location. You can therefore choose from a range of additional options to suit your individual project circumstances.
- **Triple certification:** we set benchmarks when it comes to occupational health and safety, environmental management and quality management – as evident from our certification (OHSAS 18001, ISO 14001 and ISO 9001).

The Senvion MM series.

The reliable bestseller in our onshore portfolio.

The MM series is the bestseller in our onshore portfolio. With more than 4,500 turbines in operation, these models have become the standard choice in the 2-MW-class. Hardly surprising, as this model can be used anywhere in the world, even in extreme climatic conditions, thanks to alternative versions for hot and cold climates. Three types of turbine with different rotor sizes and a choice of hub heights guarantee full flexibility and economic efficiency in all wind speeds and in any location conditions.

Senvion technology – tried-and-tested, integrated and efficient

The technology concept of the MM turbines is based on our 1.5 megawatt MD series. The MM series is equipped with a variable speed generator/frequency converter system and DC blade adjustment (pitch system) and has proven its reliability for years. The heart of our wind turbines is the integrated drive train concept, where all the different components – rotor blade, planetary/spur gear, generator and transformer – work in perfect harmony. This ideal interaction of the individual components delivers a high energy yield and also keeps noise emission low.

Smart logistics and service concepts

Apart from the design of individual components, simple and safe transport and also fast assembly all play a key role for the economic efficiency of a wind turbine. Because we use standard components, we can guarantee that they will still be available after many years. This also enables inspections to be performed safely and efficiently. Large components can be transported by ship and HGV to destinations all over the world. It goes without saying that we also keep stocks of spare parts available at diverse locations for quick and reliable supply to wind farms.

Modular systems for maximum flexibility.

MM82

MM82 – the classic turbine for high wind speeds.

With a rotor surface of 5,281 m², this 2.05 megawatt wind turbine generates consistently good energy yields in high wind locations. It is based on the technology of the MM70, which was launched in 2002 and is meanwhile one of the most frequently produced types of wind turbine in the world. Hub heights ranging from 59 to 80 metres make the MM82 the ideal choice for sites with height restrictions.

MM92

MM92 – our bestseller for medium and low wind speeds.

The Senvion MM92 has a rated power output of 2.05 megawatts at a wind speed of 12.5 m/s, thanks to its significantly larger rotor diameter of 92.5 metres. This makes it the ideal turbine for medium to low wind speeds. The MM92 produces energy so powerfully, efficiently and reliably that it has become our top-selling turbine. It is based on the dependable generator gearbox technology of the MM82.

MM100

MM100 – turns low wind speeds into revenues.

The MM100 is ideally equipped to make efficient use of all locations with low wind speeds: a rotor diameter of 100 m, a swept rotor area of 7,854 m² and hub heights ranging from 75 to 100 metres. Moreover, its low sound power level of 103.8 dB(A) makes it one of the quietest systems in its class and therefore particularly interesting for onshore wind farms which are subject to noise restrictions.







1 Generator/frequency converter

- Variable speed range for optimum yields
- Fully encapsulated generator with air-air heat exchanger
- Optimum temperature inside generator, even at high ambient temperatures

2 Gear system

- Planetary/spur gear
- Satisfies stringent demands in terms of service life and operating smoothness in line with Senvion gear specifications
- Low temperature thanks to efficient oil cooling system
- Three-stage oil filter system for excellent oil quality

3 Rotor blade

- Individual blade profiles for optimal aerodynamics of the rotor
- Robust aerodynamic design guarantees best possible performance, even in difficult wind conditions or when dirty
- Proven quality, tested statically and dynamically in accordance with IEC specifications
- Fully integrated lightning protection in accordance with IEC protection class I

4 Rotor hub

- Dimensionally stable thanks to compact design and alignment with transfer of torque
- Optimum integration of pitch drive systems
- Generously proportioned spinner ensures safe access to hub in all weathers

5 Pitch system

- Virtually maintenance free electrical system
- High-quality, large blade bearing with permanent track lubrication and external gearing
- Integrated deflector in the spinner as protection from the elements
- Maximum reliability through redundant blade angle detection with two independent measuring systems
- Fail-safe design with independent drive and control systems for each rotor blade

6 Rotor bearing and shaft

- High quality spherical roller bearing with bearing with perfectly fitted bearing housing and permanent lubrication for maximum service life
- Q&T steel rotor shaft cast in alignment with transfer of torque

7 Azimuth

- Four-point contact bearing with external gearing, powered by top quality high-performance gear motors
- Holding brakes with hydraulic pressure accumulator for fail-safe function relieve drive systems when inoperative and stabilise nacelle
- Minimum stress on the drive trains thanks to low friction of four-point bearing and brake release during wind tracking

8 Holding brake

- Generously dimensioned disc brakes hold rotor safely
- Soft brake function for minimum wear of gearbox

9 Tubular steel towers

- Natural frequency of tower higher than rotational speed of rotor (rigid design) ensures minimum stress on tower and nacelle
- No speed restrictions, as no frequency interference
- Maximum component safety thanks to L-shaped flange and stress-optimised door opening

Bus-bar (in the tower)

- Prevents electrical interference in the turbine
- Complies with VDE regulations (German Association for Electrical, Electronic and Information Technologies)
- Best possible protection in case of short circuit or fire

Lightning protection

- Internal and external lightning protection zone concept in compliance with IEC norm
- External lightning protection via blade receptors and air terminal rod on the weather vane
- Reliable protection of bearings via defined lightning paths
- GRP coupler for galvanic isolation of generator system from gearbox
- Surge protector for electrical system
- Insulated bearing shells reliably protect generator

MM top technology in detail.

SCADA Solutions-System



SCADA Solutions tell you exactly what's going on in your turbine.

To enable comprehensive monitoring and efficient control of your wind turbine or wind farm, we provide you with a customised package of software and hardware products. Our SCADA Solutions System (Supervisory Control and Data Acquisition) involves highly diverse applications, which can be combined to form exactly the package you need. SCADA also stands for user-friendliness, high-level security and easy system integration. The benefit for you is that it also ensures a high level of operating hours throughout the entire service life of the turbines.

SCADA Access is a full-service application for web-based, encrypted access to your wind farm or to individual turbines. The connection of different types of turbines and software systems to a central superordinate SCADA system is provided by our interfaces and communication products, which enable direct data transmission via OPC DA, FTPS and IEC 61400-25 protocols. Our premium application, Windfarm Management, is a powerful SCADA system for wind farms of up to 250 wind turbines.

Individual service for your wind turbine.

We want you to enjoy reliable, long-term yields – which are both flexible and calculable. Accordingly, we offer a range of different service packages to match your individual requirements, in the same way as our turbine solutions for specific projects. You can choose between a modular standard agreement and a full-maintenance contract (Integrated Service Package) with either a time-based availability warranty or a yield-based warranty*. Contracts which run for up to 20 years in combination with more than two decades of maintenance experience give your investment the reliability you need. Our service is fast, comprehensive and effective: around 700 fully qualified employees who undergo regular advanced training attend to wind turbines with a combined total output of more than 8 gigawatts. A convincing argument not only for our customers, but also for their banks.

*The production warranty under the full-maintenance contract is only available for wind farms with a minimum of five wind turbines and is subject to specific project conditions.

Why our products match your requirements.



"Our MM series sets standards in the 2MW market, particularly in terms of noise emission. No wonder it has been our bestseller for so many years."

Julia Timm,
Sales Manager, Servion Hub Europe South West



- **We satisfy the grid conditions in all major markets.**

When connecting your wind turbines to the grid, we can draw on the experience acquired from installation of more than 7,317 turbines worldwide, in both the onshore and offshore sectors. Simulations and layout recommendations at an early stage of the design process allow site-specific integration of your wind turbine into the grid. There are both 50 and 60 Hz versions available.

- **Our turbines have proved their efficiency and reliability.**

We have been proving the efficiency and reliability of our turbines in daily operations for more than 25 years. That is why we can give contractual warranties of compliance with the performance curves, technical availability and site-specific noise levels. If you are interested, we can gladly inform you of the results calculated by external advisor DNV GL for our performance curves.

- **Your investment pays off – every step of the way.**

In keeping with our principle of "Innovation through Evolution", all Senvion turbines are based on tried-and-tested platforms which have been systematically modified and enhanced. This means that every turbine contains more than 25 years of experience in the wind energy business. So it comes as no surprise that almost all of the 4,500 MM turbines we have installed worldwide are still operating with their first gearbox. This long service life is the result of extensive validation of the prototypes, regular quality control during the production process and monitoring of the turbines while in operation.

- **Our turbines can be adjusted to suit all wind classes and site conditions.**

We have special versions for very hot and very cold climate zones and a large range of models for different wind speeds. This allows you to respond flexibly to suit any site conditions. With a choice of control functions and grid connection options, such as frequency control for our 50 and 60 Hz versions or enhanced reactive power capacity, we enable grid connection in any project conditions.

- **We connect you to any customer control centre.**

Our SCADA Solutions System (Supervisory Control and Data Acquisition) consists of high-performance software and hardware products, including various interfaces and communications applications which facilitate transmission of the operating data from your wind turbine to our customer control centre at the wind farm or alternatively to any external institution.

- **Our turbines permit easy maintenance – right from the start.**

Shorter downtimes and lower financial risks – these are the targets we have in mind right at the start of the development process. Our turbines are designed for ease of maintenance so that the service engineers can carry out inspections and maintenance work on the individual components reliably and without risks, thus ensuring maximum safety.

Our references





Location: Moimenta da Beira, Vila Nova de Pava, Sernancelhe and Lousã, Portugal



WTG type: 72 x MM92, 12 x MM100



Total output: 171,6 MW



Operator: Âncora Wind Energia Eólica SA

The Âncora wind farms, Portugal

In 2016, Senvion completed the delivery and installation of 84 wind turbines for five wind farms in Portugal for its partner Âncora Wind. With its medium to low wind speeds, the region of Âncora is a prime spot for the project. However, it is also an important habitat for the last of the Iberian wolves. Because of regulatory constraints, three of the five wind farms were connected to a unique substation in Moimenta which is several kilometers away from the other wind farms. Senvion worked closely with all the involved parties and erected all turbines within 21 months, respecting both the planning schedule as well as local wildlife and their breeding season.

In addition, artists Vhils and Joana Vasconcelos created a stunning, larger-than-life artwork for the turbines. Today the project not only contributes to the production of clean and affordable energy in Portugal, but also to sustainable local job creation.



Our references



Location: Approximately 12 km north-west of Brora, in Sutherland, Scotland



WTG type: 35 x MM82



Total output: 71.75 MW



Production: 60,000 homes*



Operator: SSE



Location: Steiermark, Austria



WTG type: 9 x MM92



Total output: 18.45 MW



Production: 41,000,000 kWh per year



Operator: Windheimat GmbH



Link: www.windheimat.at

Gordonbush, Scotland

On 1st March 2012, the last of a total of 35 turbines was installed at Gordonbush, the biggest UK onshore wind farm of Senvion to date. The project has been a massive challenge for the installation team: All 35 turbines were installed during winter, in the far North of Scotland. The site has been operational since June 2012.

*based on OFGEM's typical annual domestic household consumption of 3,200 kWh a year

Hochpürschting Wind Farm, Austria

This project was commissioned in 2013, the building and erection phase having extended over two summer periods. Because it is situated at 1,500 meters above sea level it can only be supplied using automatically controlled special equipment in the summer months. The site also places high demands on operation and service, which Senvion has fulfilled by designing procedures specially for the project.



Location: South Gippsland, Victoria, Australia



WTG type: 52 x MM92



Total output: 106.6 MW



Operator: Mitsui & Co. (Australia) Ltd

Bald Hills Wind Farm, Australia

The Bald Hills Wind Farm is Senvion's largest EPC project in Australia so far and has been completed in 2015. Senvion was responsible for supplying the turbines as well as logistics, cranes, installation, commissioning and testing the WECs together with the electrical and civil balance of the plant. That also included design and construction of the substation, underground cable reticulation, met masts, gravel roads and crane hardstands and turbine foundations.

"Gayton le Marsh wind farm is our fifth contract with Senvion and demonstrates our confidence in the turbines and service package. We look forward to working with Senvion's team to combine our expertise and ensure the project is built to the highest standards and completed on time."

Richard Lill,
Financing and Construction Manager
of Energiekontor UK





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